Lecture 13: Maps

Heather Mattie, PhD

October 13, 2026

Recipe of the Day!

Butternut Squash Pasta with Walnuts and Sage





Connors Farm
Danvers, MA



Announcements

- Homework #3 due Friday, October 23 by 11:59pm
- Heather's office hour: Thursday 12-1pm
- No lab this week



Coding question of the day

Using the **`gapminder`** dataset, make a line graph of life expectancy over time. Include one line per country, and another line with the global average life expectancy. Be sure to label the global average line and to make it stick out among the country-specific lines. Use the code below to calculate the crude global average for each year.

Hints:

1. Use alpha to help make the global average line stick out.
2. Use 2 **`geom_line`** layers.
3. Note that **`global_avg`** is a data frame.

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data(gapminder)
```

```
global_avg <- gapminder %>%
  group_by(year) %>%
  summarize(life_expectancy = mean(life_expectancy, na.rm = TRUE)) %>%
  ungroup()
```

Maps

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Maps Roadmap

1. The `maps` package
2. Plotting global life expectancy
3. Plotting US state murder rates
4. Case study: COVID-19 cases

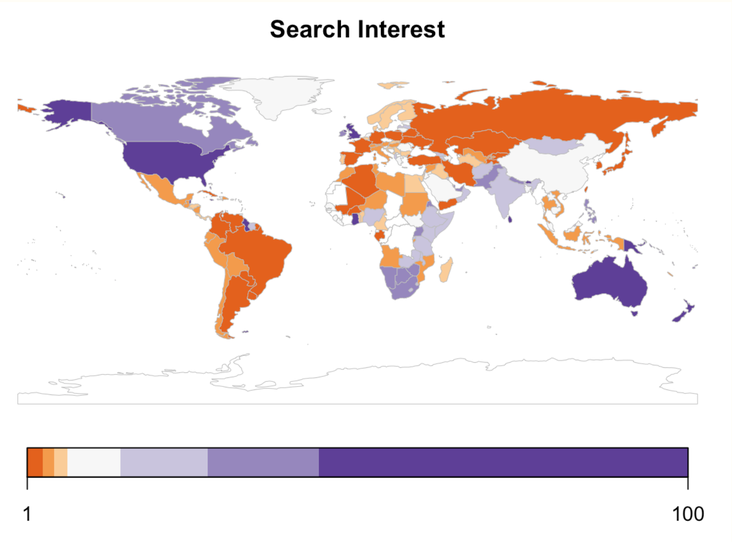
Maps

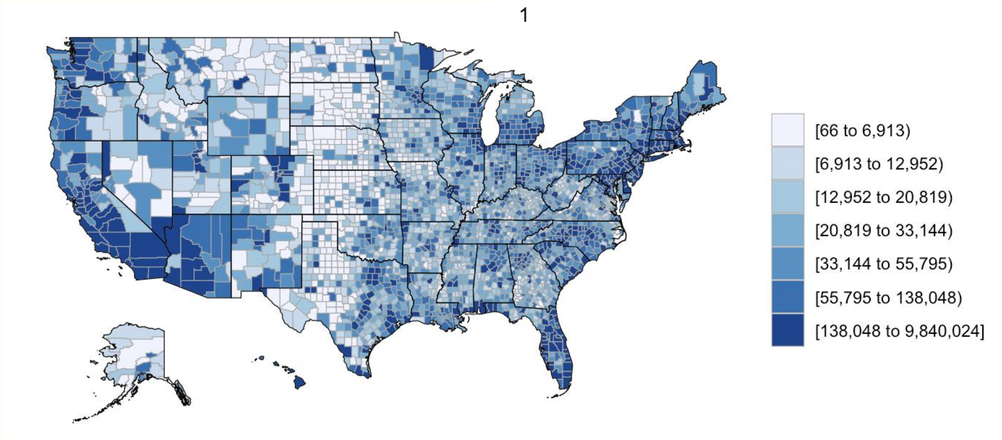
The `maps` package

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Maps

- Plots of maps can be very powerful, informative and aesthetically pleasing
- They can also be misleading if not created correctly
- We will us the **`maps`** package to create plots of maps
- Other packages: **`ggspatial`**, **`sf`**, **`urbnmapr`**, **`rnaturalearth`**, etc.

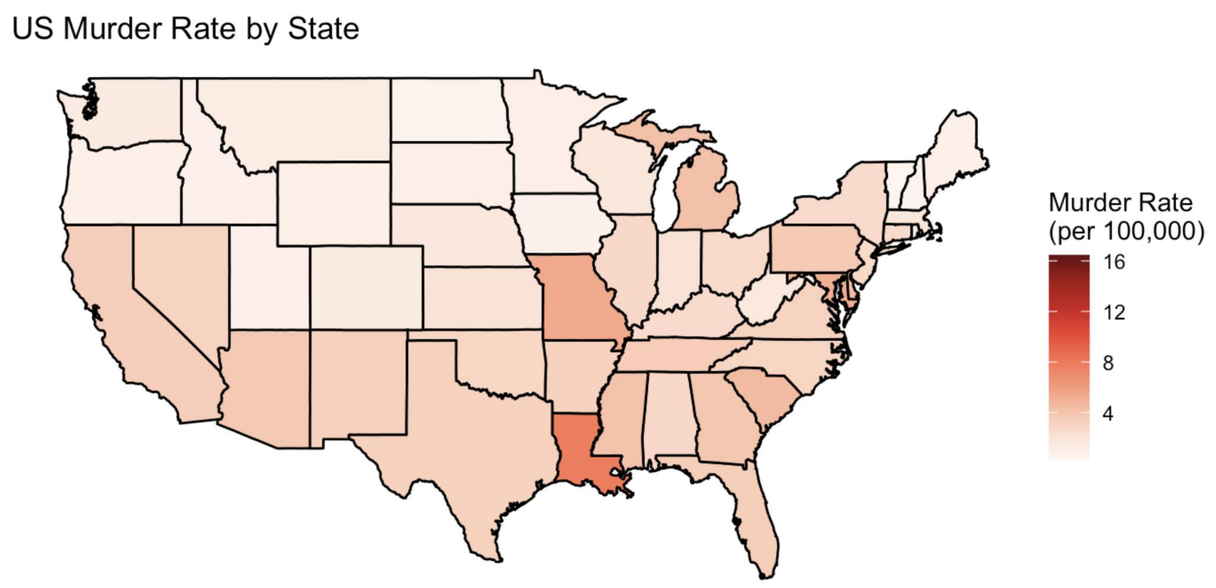




maps

The maps package contains outlines of continents, countries, states, and counties / jurisdictions / provinces

- Specify which map to use with `map_data()`
- Loaded as a data frame and can be used with the pipe (`%>%`) and `ggplot`



map_data("world")

- **long** = longitude
- **lat** = latitude
- **group** indicates which countries can be drawn together
- **order** indicates which order to connect the **lat** and **long** dots
- **region** = country
- **subregion** = county, political jurisdictions, etc.

```
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

```
##        long      lat group order region subregion
## 1 -69.89912 12.45200     1     1  Aruba      <NA>
## 2 -69.89571 12.42300     1     2  Aruba      <NA>
## 3 -69.94219 12.43853     1     3  Aruba      <NA>
## 4 -70.00415 12.50049     1     4  Aruba      <NA>
## 5 -70.06612 12.54697     1     5  Aruba      <NA>
## 6 -70.05088 12.59707     1     6  Aruba      <NA>
```

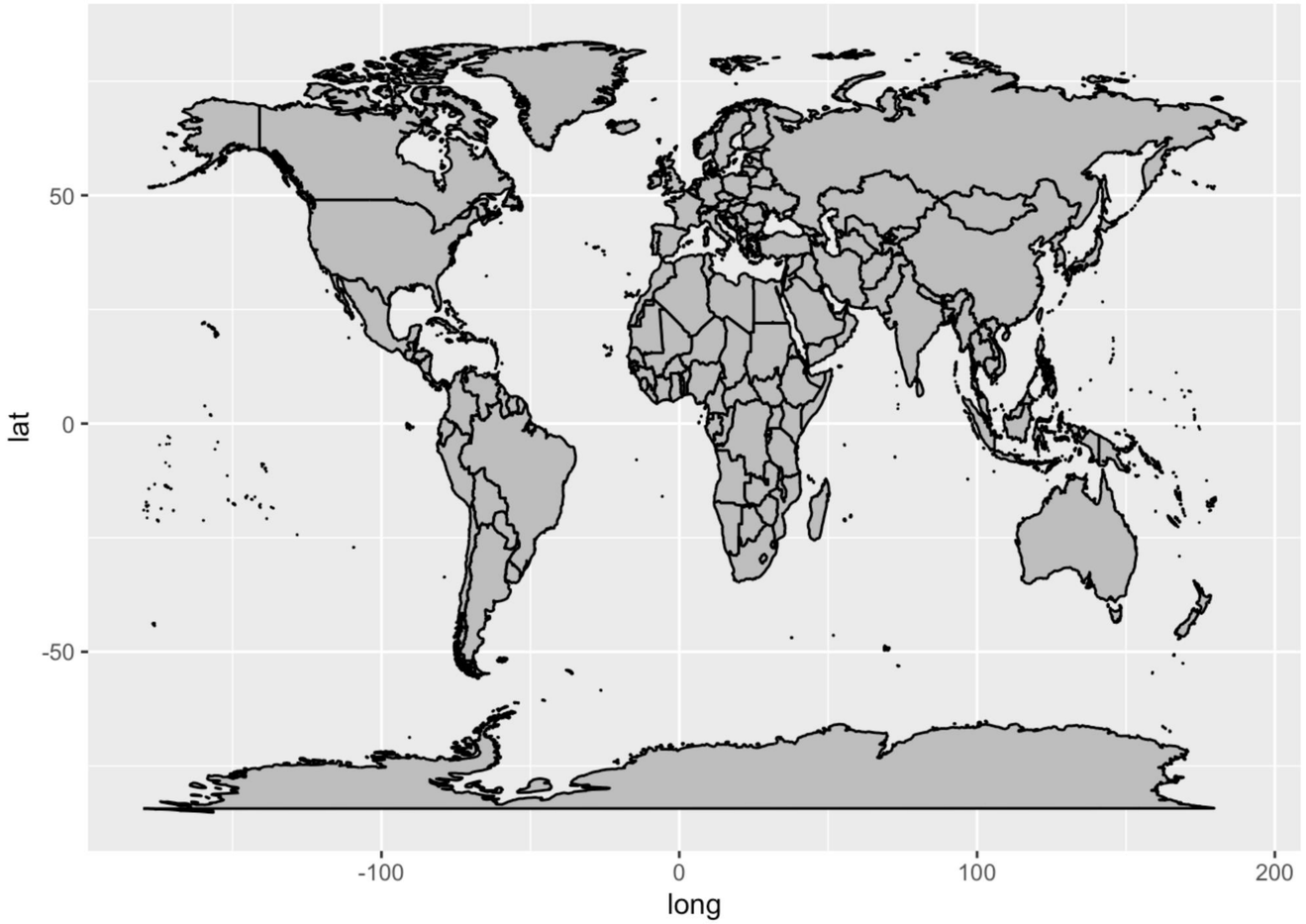
```
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```

map_data("world")

```
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

```
##        long      lat group order region subregion
## 1 -69.89912 12.45200     1     1  Aruba      <NA>
## 2 -69.89571 12.42300     1     2  Aruba      <NA>
## 3 -69.94219 12.43853     1     3  Aruba      <NA>
## 4 -70.00415 12.50049     1     4  Aruba      <NA>
## 5 -70.06612 12.54697     1     5  Aruba      <NA>
## 6 -70.05088 12.59707     1     6  Aruba      <NA>
```

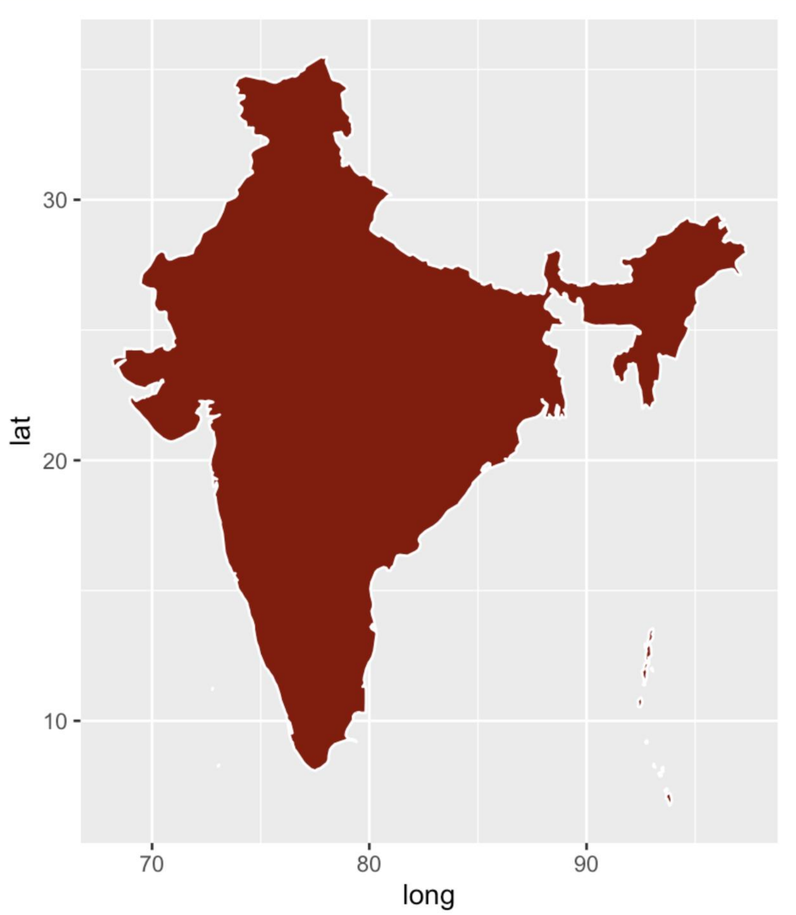
```
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```



Countries

```
india_map <- map_data("world", "India")

india_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.2)
```

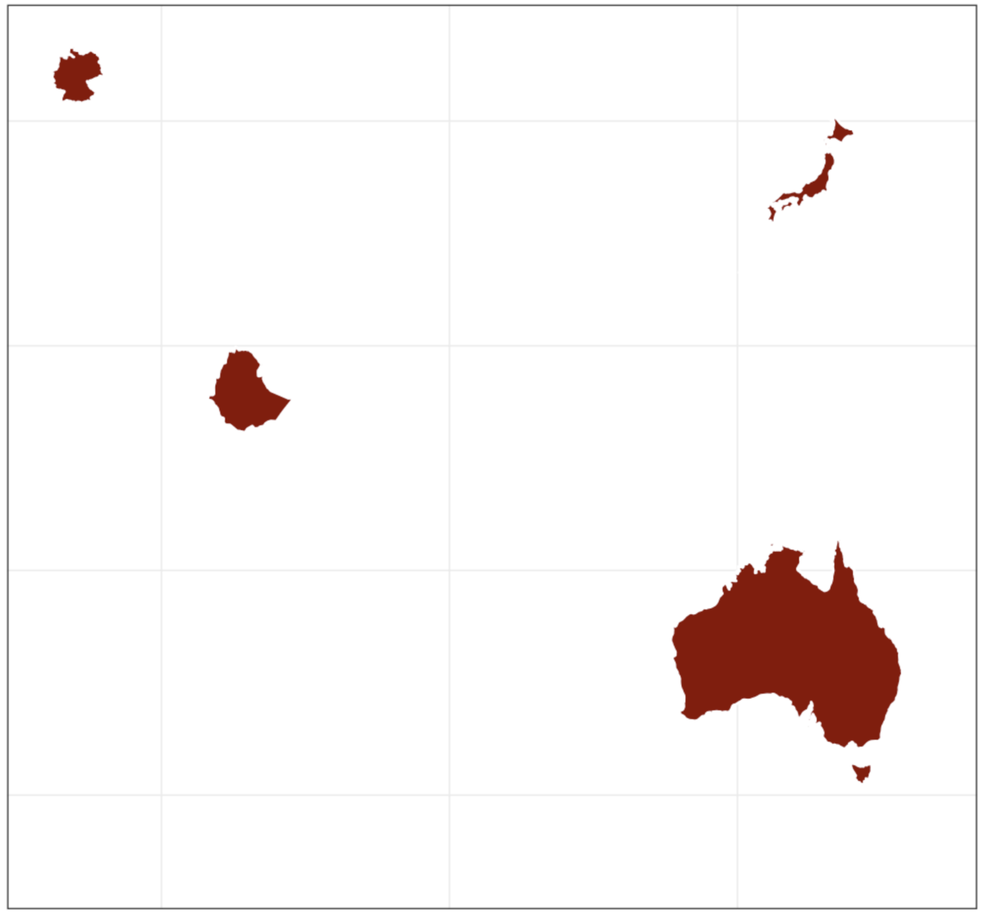


Countries

```
theme_set(theme_bw())

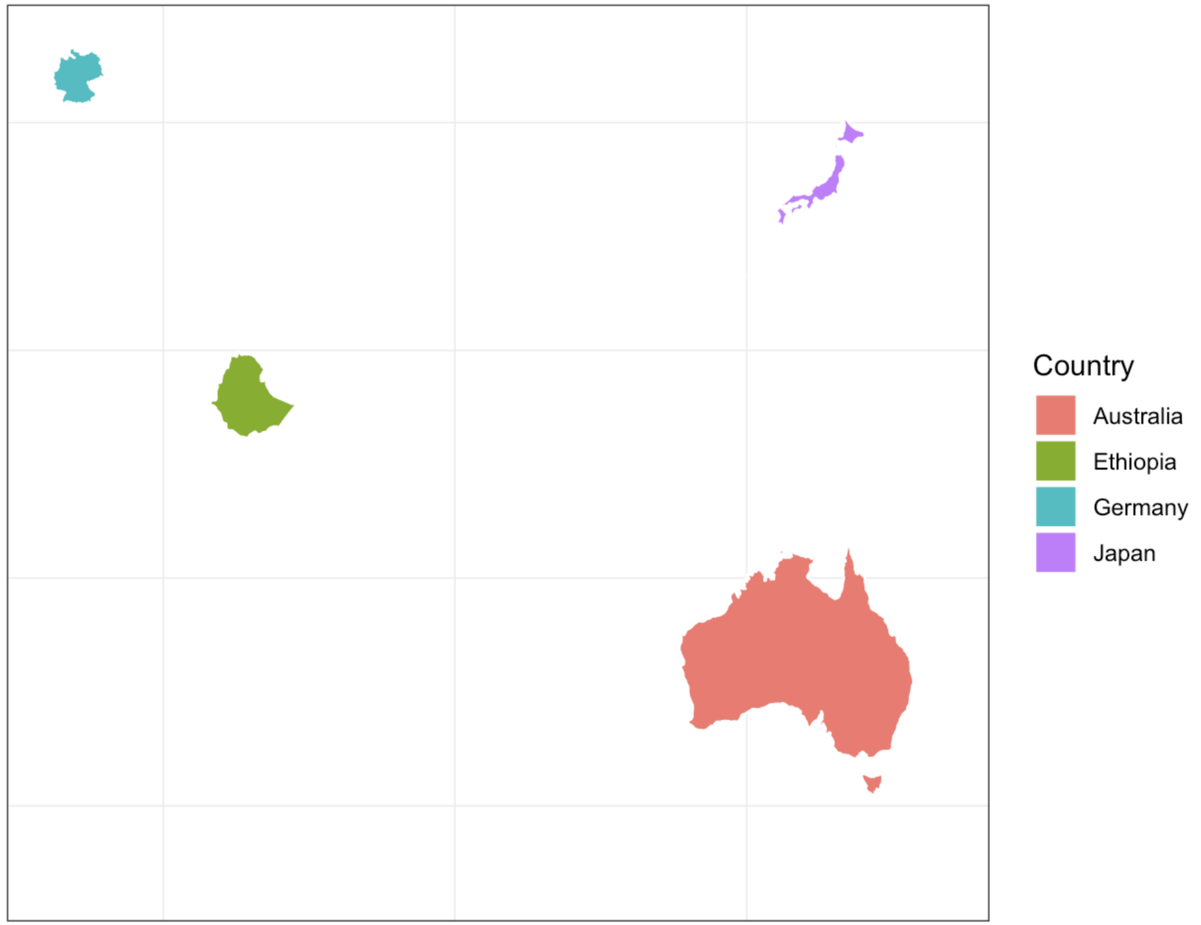
multiple_countries_map <- map_data("world",
                                   c("Japan", "Ethiopia", "Australia", "Germany"))

multiple_countries_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  theme(panel.grid.major = element_blank(),
  panel.background = element_blank(),
  axis.title = element_blank(),
  axis.text = element_blank(),
  axis.ticks = element_blank()) +
  coord_fixed(1.3)
```



Countries

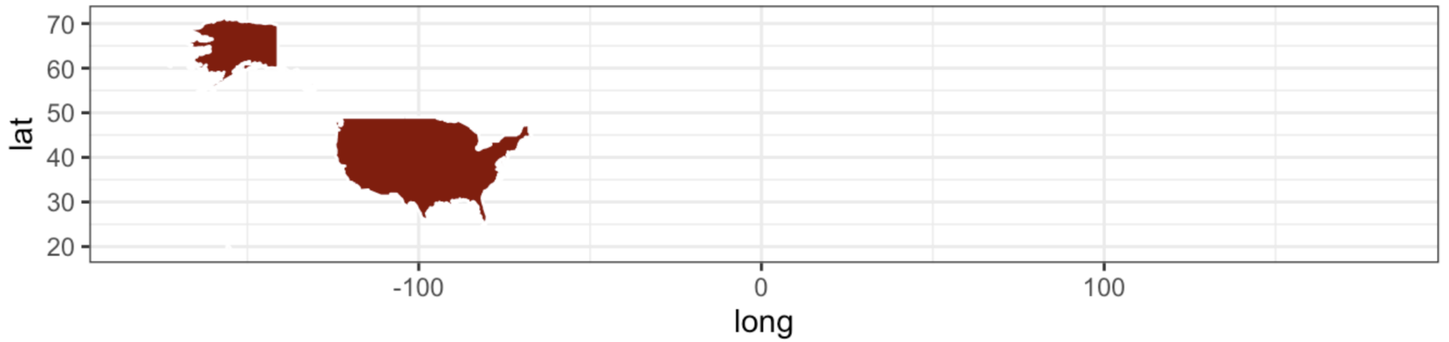
```
multiple_countries_map %>%
  ggplot(aes(x = long, y = lat, group = group, fill = region)) +
  geom_polygon(color = "white") +
  theme(panel.grid.major = element_blank(),
  panel.background = element_blank(),
  axis.title = element_blank(),
  axis.text = element_blank(),
  axis.ticks = element_blank()) +
  labs(fill = "Country") +
  coord_fixed(1.3)
```



US maps

```
us_map <- map_data("world", "usa")

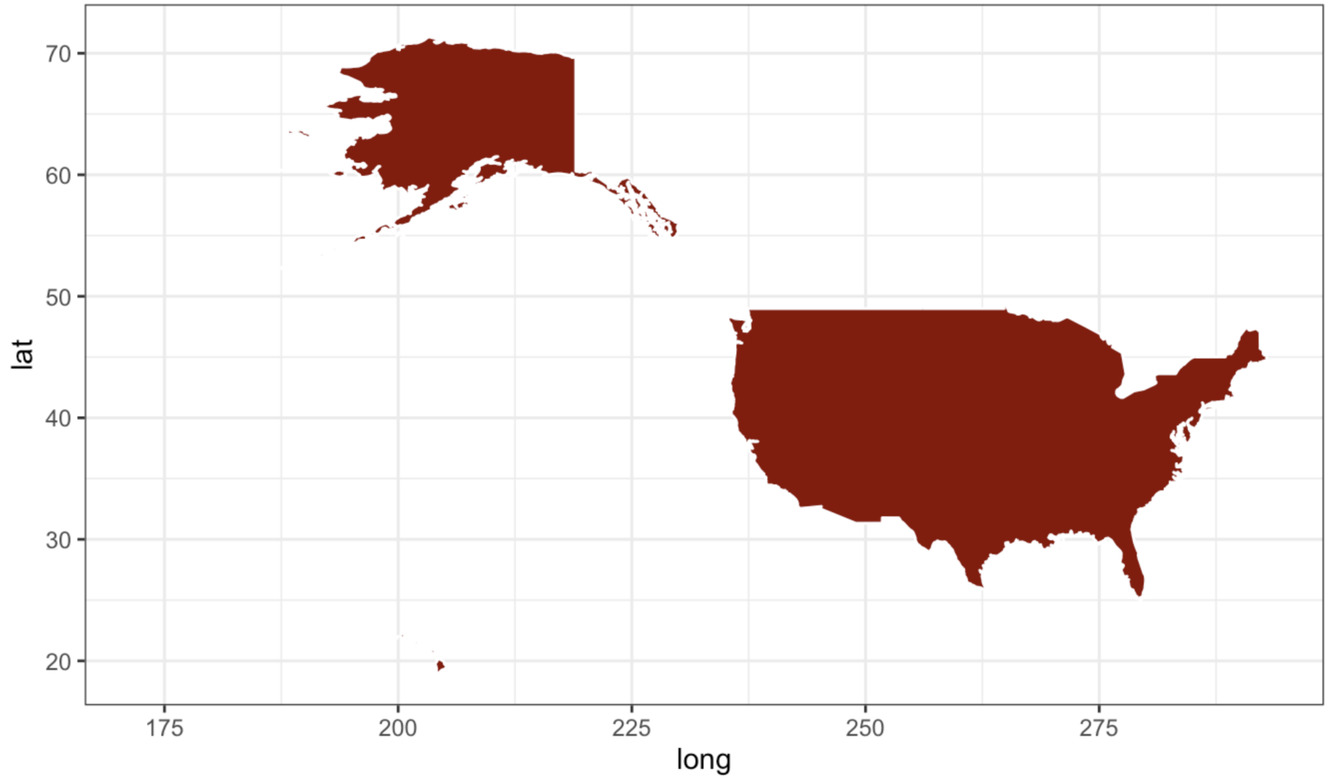
us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



US maps

```
better_us_map <- map_data("world2", "USA")

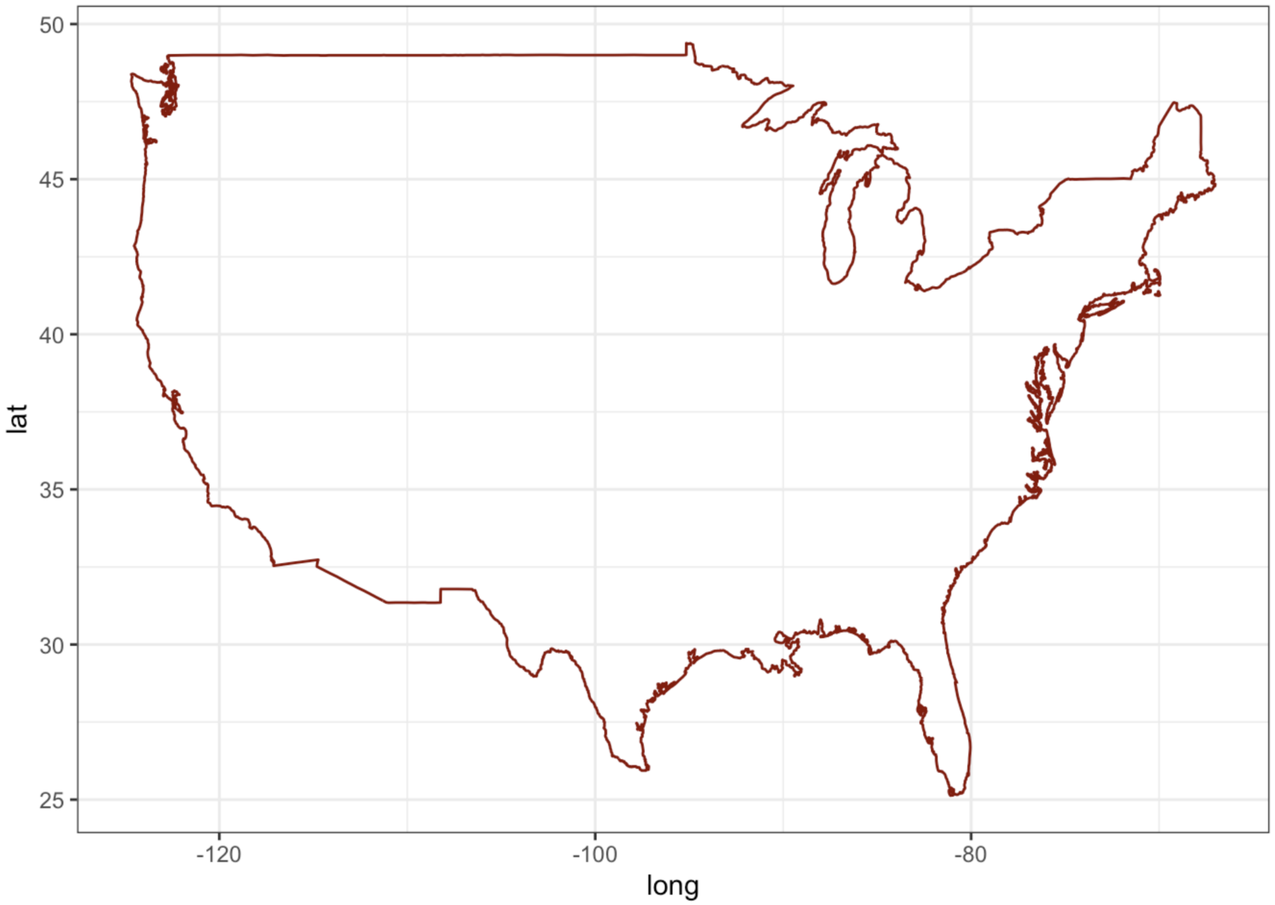
better_us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



US maps

```
usa <- map_data("usa")

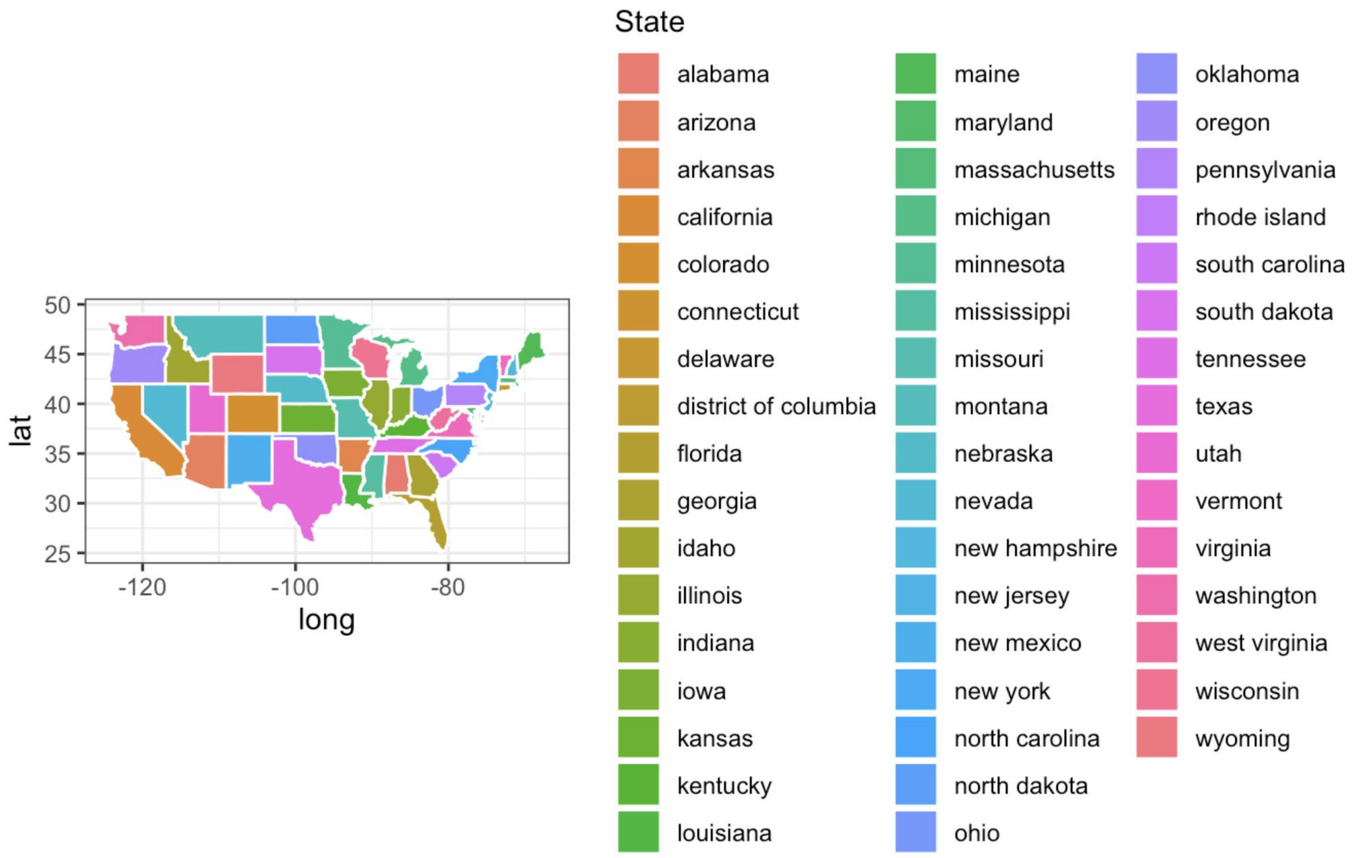
usa %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA)
```



US states

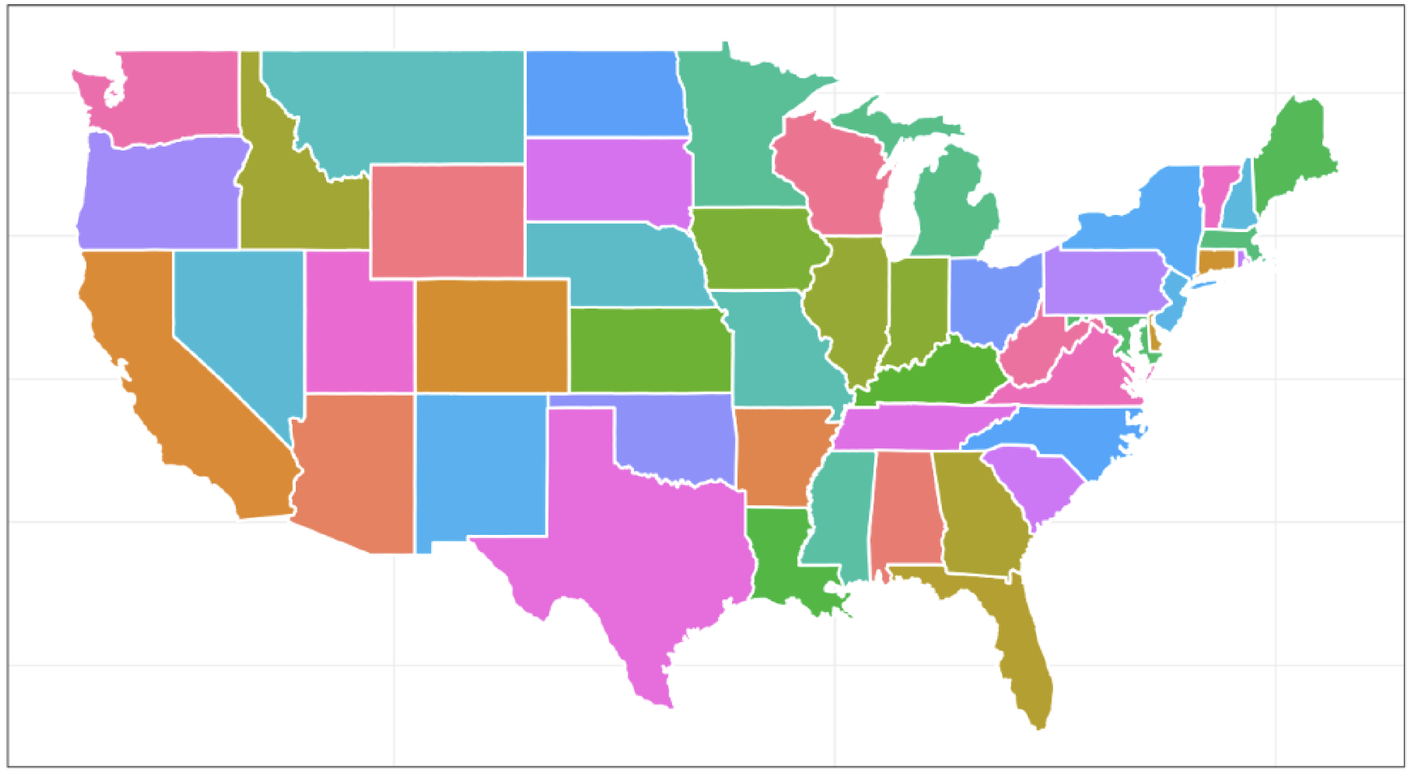
```
us_states <- map_data("state")

us_states %>%
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +
  geom_polygon(color = "white") +
  labs(fill = "State") +
  coord_fixed(1.3)
```



US states

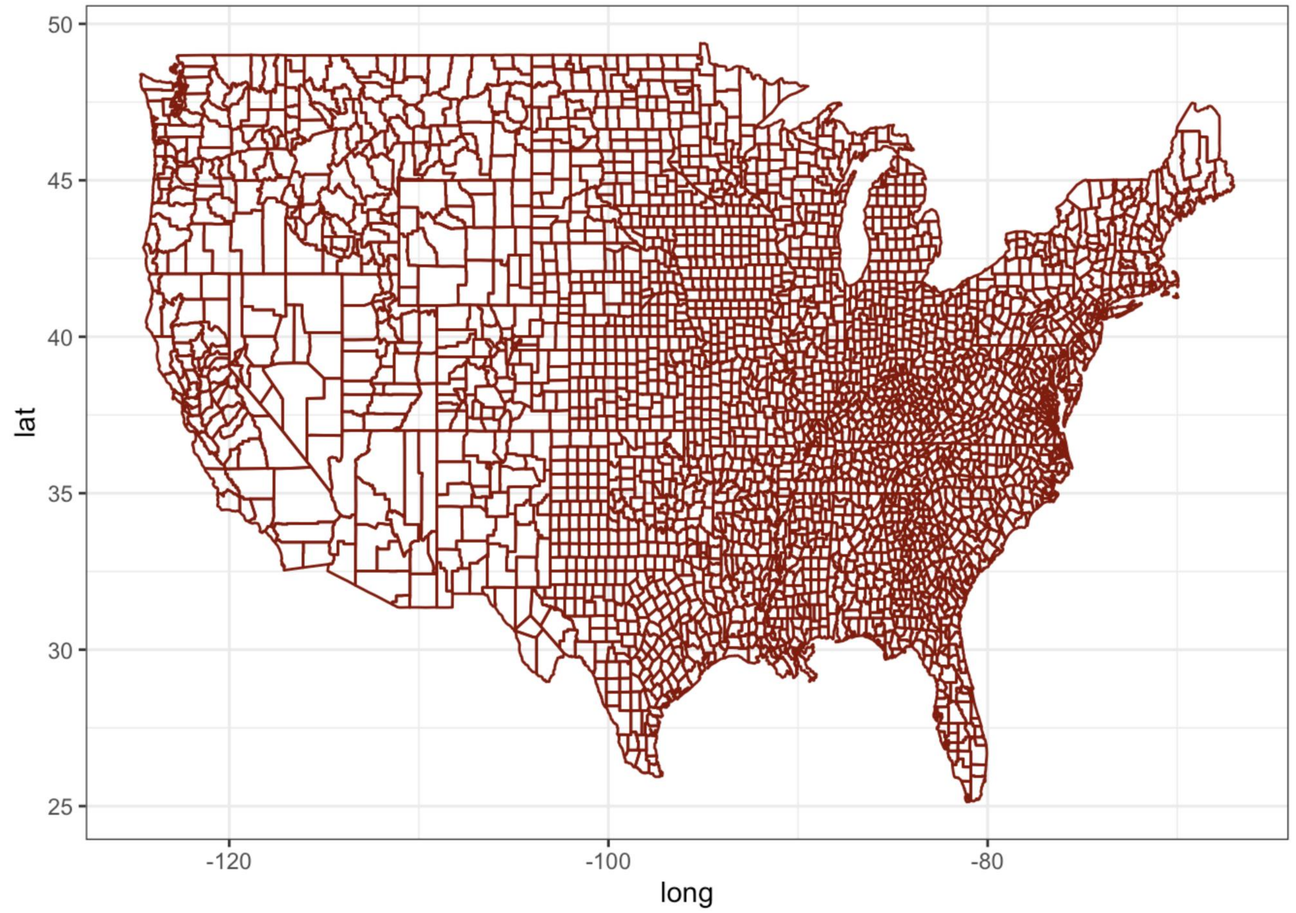
```
us_states %>%
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +
  geom_polygon(color = "white") +
  theme(panel.grid.major = element_blank(),
  panel.background = element_blank(),
  axis.title = element_blank(),
  axis.text = element_blank(),
  axis.ticks = element_blank()) +
  coord_fixed(1.3) +
  guides(fill = "none")
```



US counties

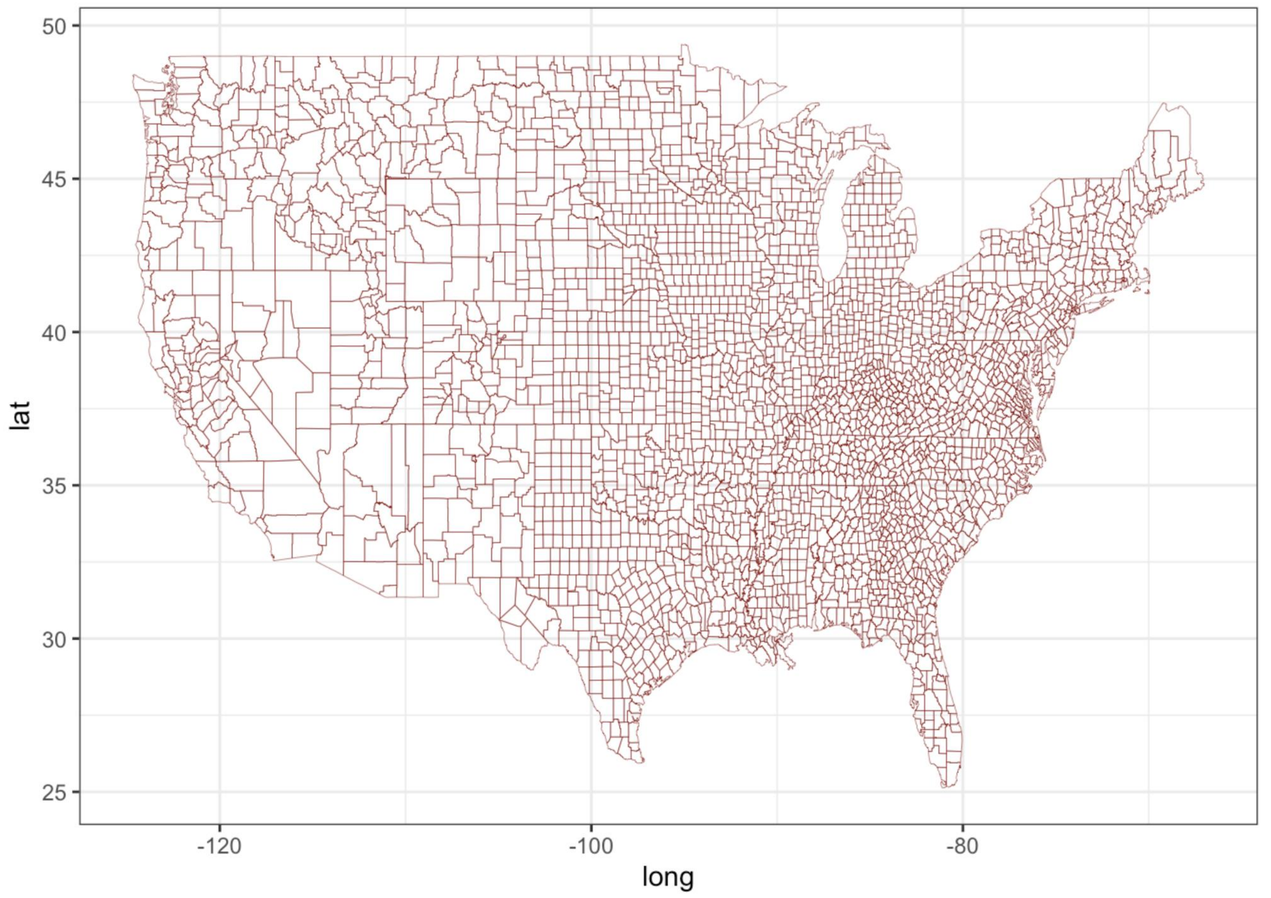
```
AllCounty <- map_data("county")

AllCounty %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA)
```



US counties

```
AllCounty %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA, linewidth = .1)
```



Summary

- The `maps` package can be used to produce maps of continents, countries, states, counties, and provinces
- The `map_data()` function loads map data
- Color or labels should be used instead of a legend if there are too many categories (countries, states, etc.) represented in the map

Maps

Plotting global life expectancy

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Plotting global life expectancy

Task: plot a map displaying the life expectancy in 2015 for countries with data in the **`gapminder`** dataset

```
library(plyr)
library(tidyverse)
library(maps)
library(dslabs)

data(gapminder)
gapminder_2015 <- gapminder %>%
  filter(year == 2015)

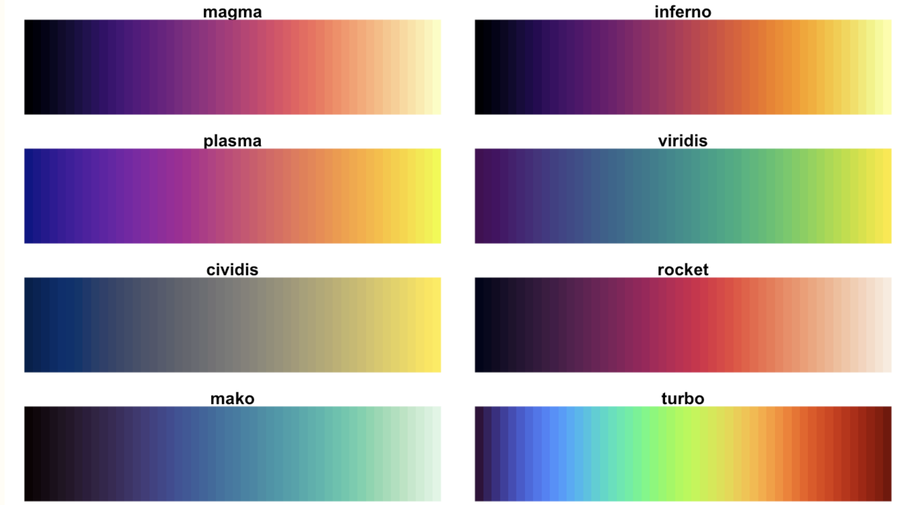
world_map <- map_data("world")
merged_world <- left_join(world_map, gapminder_2015,
                          by = c("region" = "country"))

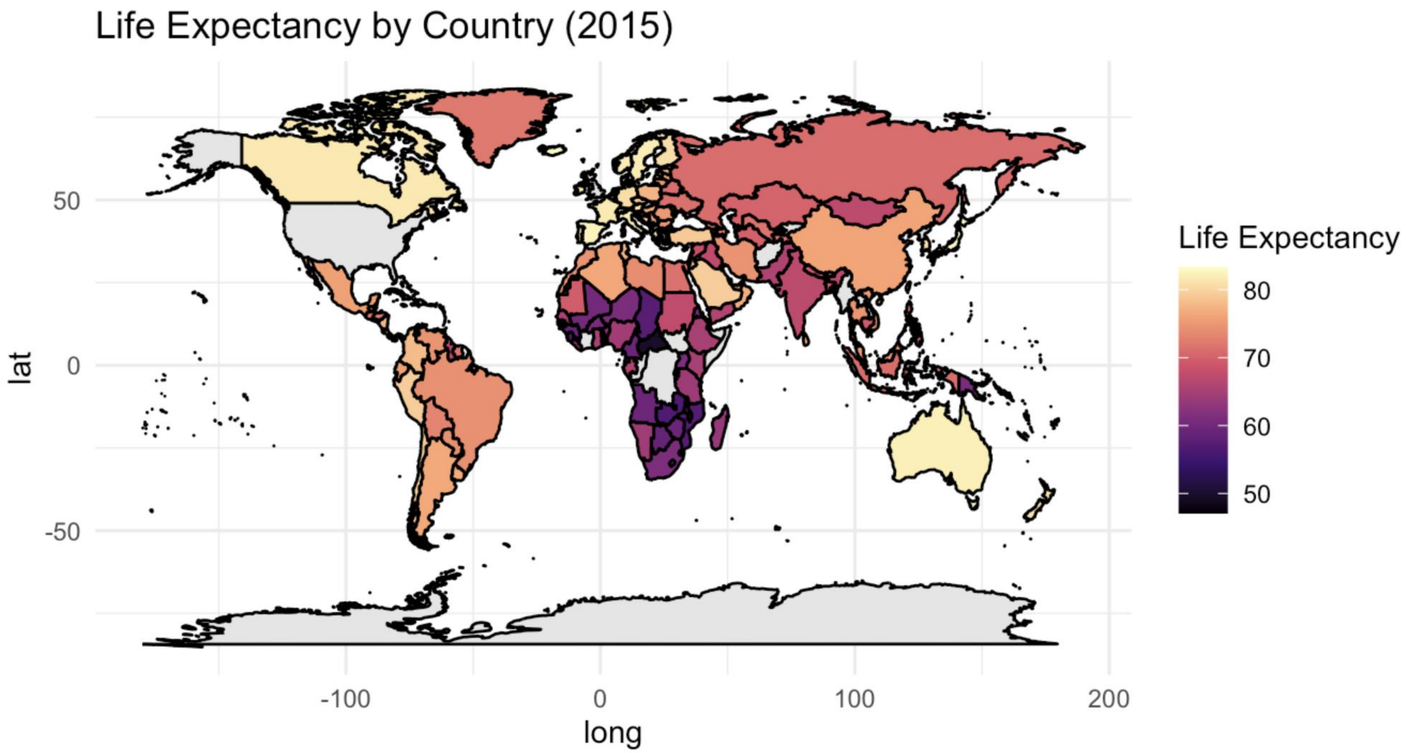
head(merged_world)
```

```
##        long      lat group order region subregion year infant_mortality
## 1 -69.89912 12.45200     1     1  Aruba      <NA> 2015               NA
## 2 -69.89571 12.42300     1     2  Aruba      <NA> 2015               NA
## 3 -69.94219 12.43853     1     3  Aruba      <NA> 2015               NA
## 4 -70.00415 12.50049     1     4  Aruba      <NA> 2015               NA
## 5 -70.06612 12.54697     1     5  Aruba      <NA> 2015               NA
## 6 -70.05088 12.59707     1     6  Aruba      <NA> 2015               NA
##   life_expectancy fertility population gdp continent  region.y
## 1           75.72      1.66     103889  NA  Americas Caribbean
## 2           75.72      1.66     103889  NA  Americas Caribbean
## 3           75.72      1.66     103889  NA  Americas Caribbean
## 4           75.72      1.66     103889  NA  Americas Caribbean
## 5           75.72      1.66     103889  NA  Americas Caribbean
## 6           75.72      1.66     103889  NA  Americas Caribbean
```

Plotting global life expectancy

```
ggplot(merged_world, aes(long, lat, group = group,
                         fill = life_expectancy)) +
  geom_polygon(color = "black") +
  scale_fill_viridis_c(option = "magma",
                       na.value = "gray90") +
  theme_minimal() +
  labs(title = "Life Expectancy by Country (2015)",
       fill = "Life Expectancy") +
  coord_fixed(1.3)
```





Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
##  [1] "Afghanistan"                          "American Samoa"
##  [3] "Andorra"                              "Anguilla"
##  [5] "Antarctica"                           "Antigua"
##  [7] "Ascension Island"                     "Azores"
##  [9] "Barbuda"                              "Bermuda"
## [11] "Bonaire"                              "Canary Islands"
## [13] "Cayman Islands"                       "Chagos Archipelago"
## [15] "Christmas Island"                     "Cocos Islands"
## [17] "Cook Islands"                         "Curacao"
## [19] "Democratic Republic of the Congo"     "Dominica"
## [21] "Falkland Islands"                     "Faroe Islands"
## [23] "French Guiana"                        "French Southern and Antarctic Lands"
## [25] "Grenadines"                           "Guadeloupe"
## [27] "Guam"                                 "Guernsey"
## [29] "Heard Island"                         "Isle of Man"
## [31] "Ivory Coast"                          "Jersey"
## [33] "Kosovo"                               "Kyrgyzstan"
## [35] "Laos"                                 "Liechtenstein"
## [37] "Madeira Islands"                      "Marshall Islands"
## [39] "Martinique"                           "Mayotte"
## [41] "Micronesia"                           "Monaco"
## [43] "Montserrat"                           "Myanmar"
## [45] "Nauru"                                "Nevis"
```

```
levels(gapminder_2015$country)
```

```
##   [1] "Albania"                          "Algeria"
##   [3] "Angola"                           "Antigua and Barbuda"
##   [5] "Argentina"                        "Armenia"
##   [7] "Aruba"                            "Australia"
##   [9] "Austria"                          "Azerbaijan"
##  [11] "Bahamas"                          "Bahrain"
##  [13] "Bangladesh"                       "Barbados"
##  [15] "Belarus"                          "Belgium"
##  [17] "Belize"                           "Benin"
##  [19] "Bhutan"                           "Bolivia"
##  [21] "Bosnia and Herzegovina"           "Botswana"
##  [23] "Brazil"                           "Brunei"
##  [25] "Bulgaria"                         "Burkina Faso"
##  [27] "Burundi"                          "Cambodia"
##  [29] "Cameroon"                         "Canada"
##  [31] "Cape Verde"                       "Central African Republic"
##  [33] "Chad"                             "Chile"
##  [35] "China"                            "Colombia"
##  [37] "Comoros"                          "Congo, Dem. Rep."
##  [39] "Congo, Rep."                      "Costa Rica"
##  [41] "Cote d'Ivoire"                    "Croatia"
##  [43] "Cuba"                             "Cyprus"
##  [45] "Czech Republic"                   "Denmark"
##  [47] "Djibouti"                         "Dominican Republic"
##  [49] "Ecuador"                          "Egypt"
##  [51] "El Salvador"                      "Equatorial Guinea"
##  [53] "Eritrea"                          "Estonia"
```

Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
##  [1] "Afghanistan"                    "American Samoa"
##  [3] "Andorra"                        "Anguilla"
##  [5] "Antarctica"                     "Antigua"
##  [7] "Ascension Island"               "Azores"
##  [9] "Barbuda"                        "Bermuda"
## [11] "Bonaire"                        "Canary Islands"
## [13] "Cayman Islands"                 "Chagos Archipelago"
## [15] "Christmas Island"               "Cocos Islands"
## [17] "Cook Islands"                   "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands"               "Faroe Islands"
## [23] "French Guiana"                  "French Southern and Antarctic Lands"
## [25] "Grenadines"                     "Guadeloupe"
## [27] "Guam"                           "Guernsey"
## [29] "Heard Island"                   "Isle of Man"
## [31] "Ivory Coast"                    "Jersey"
## [33] "Kosovo"                         "Kyrgyzstan"
## [35] "Laos"                           "Liechtenstein"
## [37] "Madeira Islands"                "Marshall Islands"
## [39] "Martinique"                     "Mayotte"
## [41] "Micronesia"                     "Monaco"
## [43] "Montserrat"                     "Myanmar"
## [45] "Nauru"                          "Nevis"
```

```
levels(gapminder_2015$country)
```

```
##   [1] "Albania"                    "Algeria"
##   [3] "Angola"                     "Antigua and Barbuda"
##   [5] "Argentina"                  "Armenia"
##   [7] "Aruba"                      "Australia"
##   [9] "Austria"                    "Azerbaijan"
##  [11] "Bahamas"                    "Bahrain"
##  [13] "Bangladesh"                 "Barbados"
##  [15] "Belarus"                    "Belgium"
##  [17] "Belize"                     "Benin"
##  [19] "Bhutan"                     "Bolivia"
##  [21] "Bosnia and Herzegovina"     "Botswana"
##  [23] "Brazil"                     "Brunei"
##  [25] "Bulgaria"                   "Burkina Faso"
##  [27] "Burundi"                    "Cambodia"
##  [29] "Cameroon"                   "Canada"
##  [31] "Cape Verde"                 "Central African Republic"
##  [33] "Chad"                       "Chile"
##  [35] "China"                      "Colombia"
##  [37] "Comoros"                    "Congo, Dem. Rep."
##  [39] "Congo, Rep."                "Costa Rica"
##  [41] "Cote d'Ivoire"              "Croatia"
##  [43] "Cuba"                       "Cyprus"
##  [45] "Czech Republic"             "Denmark"
##  [47] "Djibouti"                   "Dominican Republic"
##  [49] "Ecuador"                    "Egypt"
##  [51] "El Salvador"                "Equatorial Guinea"
##  [53] "Eritrea"                    "Estonia"
```

Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
##  [1] "Afghanistan"                    "American Samoa"
##  [3] "Andorra"                        "Anguilla"
##  [5] "Antarctica"                     "Antigua"
##  [7] "Ascension Island"               "Azores"
##  [9] "Barbuda"                        "Bermuda"
## [11] "Bonaire"                        "Canary Islands"
## [13] "Cayman Islands"                 "Chagos Archipelago"
## [15] "Christmas Island"               "Cocos Islands"
## [17] "Cook Islands"                   "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands"               "Faroe Islands"
## [23] "French Guiana"                  "French Southern and Antarctic Lands"
## [25] "Grenadines"                     "Guadeloupe"
## [27] "Guam"                           "Guernsey"
## [29] "Heard Island"                   "Isle of Man"
## [31] "Ivory Coast"                    "Jersey"
## [33] "Kosovo"                         "Kyrgyzstan"
## [35] "Laos"                           "Liechtenstein"
## [37] "Madeira Islands"                "Marshall Islands"
## [39] "Martinique"                     "Mayotte"
## [41] "Micronesia"                     "Monaco"
## [43] "Montserrat"                     "Myanmar"
## [45] "Nauru"                          "Nevis"
```

```
levels(gapminder_2015$country)
```

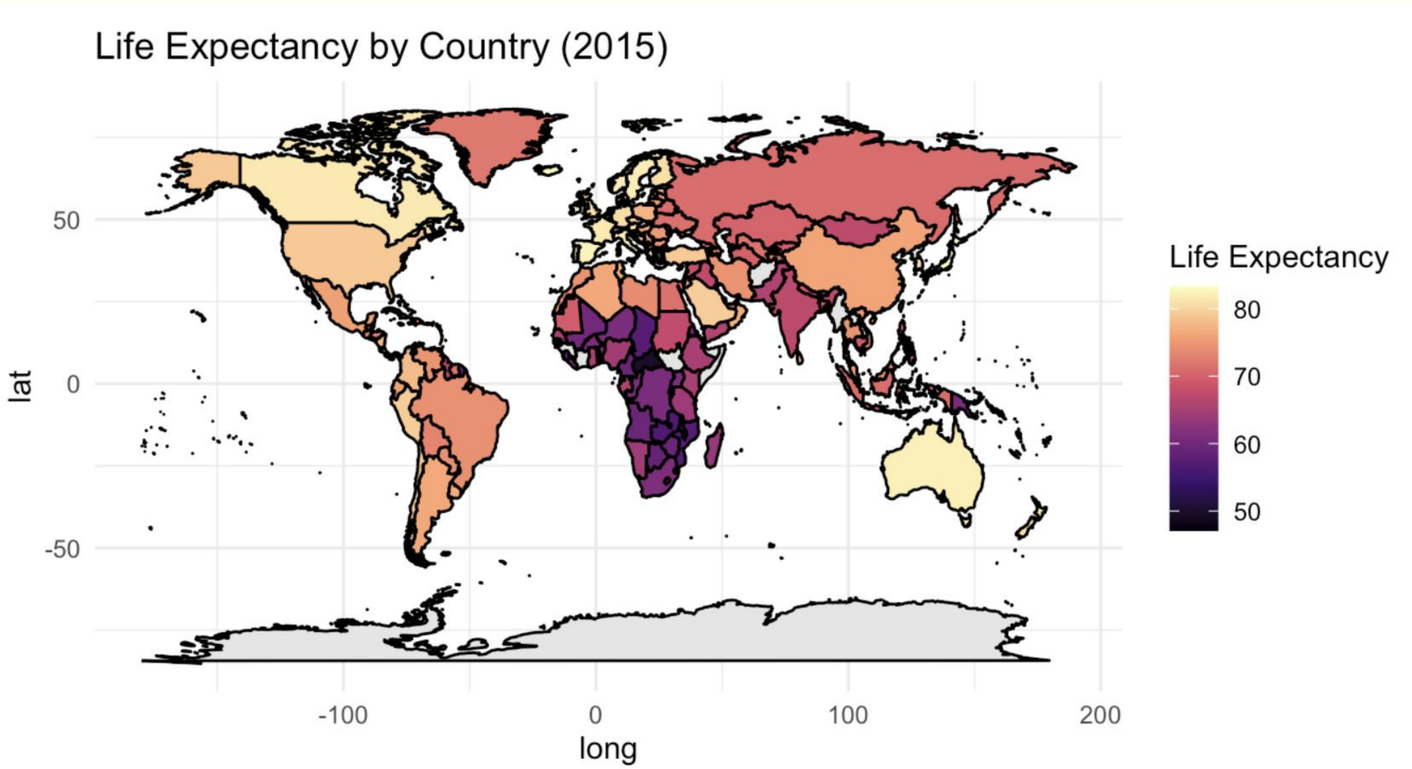
```
##   [1] "Albania"                  "Algeria"
##   [3] "Angola"                   "Antigua and Barbuda"
##   [5] "Argentina"                "Armenia"
##   [7] "Aruba"                    "Australia"
##   [9] "Austria"                  "Azerbaijan"
##  [11] "Bahamas"                  "Bahrain"
##  [13] "Bangladesh"               "Barbados"
##  [15] "Belarus"                  "Belgium"
##  [17] "Belize"                   "Benin"
##  [19] "Bhutan"                   "Bolivia"
##  [21] "Bosnia and Herzegovina"   "Botswana"
##  [23] "Brazil"                   "Brunei"
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##  [27] "Burundi"                  "Cambodia"
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##  [33] "Chad"                     "Chile"
##  [35] "China"                    "Colombia"
##  [37] "Comoros"                  "Congo, Dem. Rep."
##  [39] "Congo, Rep."              "Costa Rica"
##  [41] "Cote d'Ivoire"            "Croatia"
##  [43] "Cuba"                     "Cyprus"
##  [45] "Czech Republic"           "Denmark"
##  [47] "Djibouti"                 "Dominican Republic"
##  [49] "Ecuador"                  "Egypt"
##  [51] "El Salvador"              "Equatorial Guinea"
##  [53] "Eritrea"                  "Estonia"
```

Renaming

```
gapminder_2015 <- gapminder %>% filter(year == 2015) %>%
  mutate(country =
           revalue(country, c("Antigua and Barbuda" = "Antigua",
                              "Samoa" = "American Samoa",
                              "Congo, Rep." = "Republic of Congo",
                              "Congo, Dem. Rep." = "Democratic Republic of the Congo",
                              "Guinea" = "French Guiana",
                              "Kyrgyz Republic" = "Kyrgyzstan",
                              "St. Lucia" = "Saint Lucia",
                              "St. Vincent and the Grenadines" = "Saint Vincent",
                              "Trinidad and Tobago" = "Trinidad",
                              "United Kingdom" = "UK",
                              "United States" = "USA")))

world_map <- map_data("world")
merged_world <- left_join(world_map, gapminder_2015, by = c("region" = "country"))
```

```
ggplot(merged_world, aes(long, lat, group = group, fill = life_expectancy)) +
  geom_polygon(color = "black") +
  scale_fill_viridis_c(option = "magma", na.value = "gray90") +
  theme_minimal() +
  labs(title = "Life Expectancy by Country (2015)", fill = "Life Expectancy") +
  coord_fixed(1.3)
```



Summary

- Inconsistencies in the spelling of locations can lead to missing data in plots of maps
- Renaming the locations in one data frame before joining to a second data frame can solve a lot of inconsistencies
- Sometimes new data will need to be collected to complete a map plot

Maps

Plotting US state murder rates

Heather Mattie, PhD

Plotting state murder rates

- Task: plot a map of the US states and fill in each state with a color representing the murder rate in that state using the **`murders`** dataset
- Need to use state map data and add the murder rate for each state to the map data frame

```
library(tidyverse)
library(maps)
library(dslabs)
library(RColorBrewer)

data(murders)
murders <- murders %>% mutate(rate = total / population * 100000)
states_map <- map_data("state")

head(murders)
```

```
##        state abb region population total     rate
## 1    Alabama  AL  South    4779736   135 2.824424
## 2     Alaska  AK   West     710231    19 2.675186
## 3    Arizona  AZ   West    6392017   232 3.629527
## 4   Arkansas  AR  South    2915918    93 3.189390
## 5 California  CA   West   37253956  1257 3.374138
## 6   Colorado  CO   West    5029196    65 1.292453
```

```
head(states_map)
```

```
##        long      lat group order  region subregion
## 1 -87.46201 30.38968     1     1 alabama      <NA>
## 2 -87.48493 30.37249     1     2 alabama      <NA>
## 3 -87.52503 30.37249     1     3 alabama      <NA>
## 4 -87.53076 30.33239     1     4 alabama      <NA>
## 5 -87.57087 30.32665     1     5 alabama      <NA>
## 6 -87.58806 30.32665     1     6 alabama      <NA>
```

Plotting state murder rates

- Task: plot a map of the US states and fill in each state with a color representing the murder rate in that state using the **`murders`** dataset
- Need to use state map data and add the murder rate for each state to the map data frame

```
murders <- murders %>% mutate(region = tolower(state))
head(murders)
```

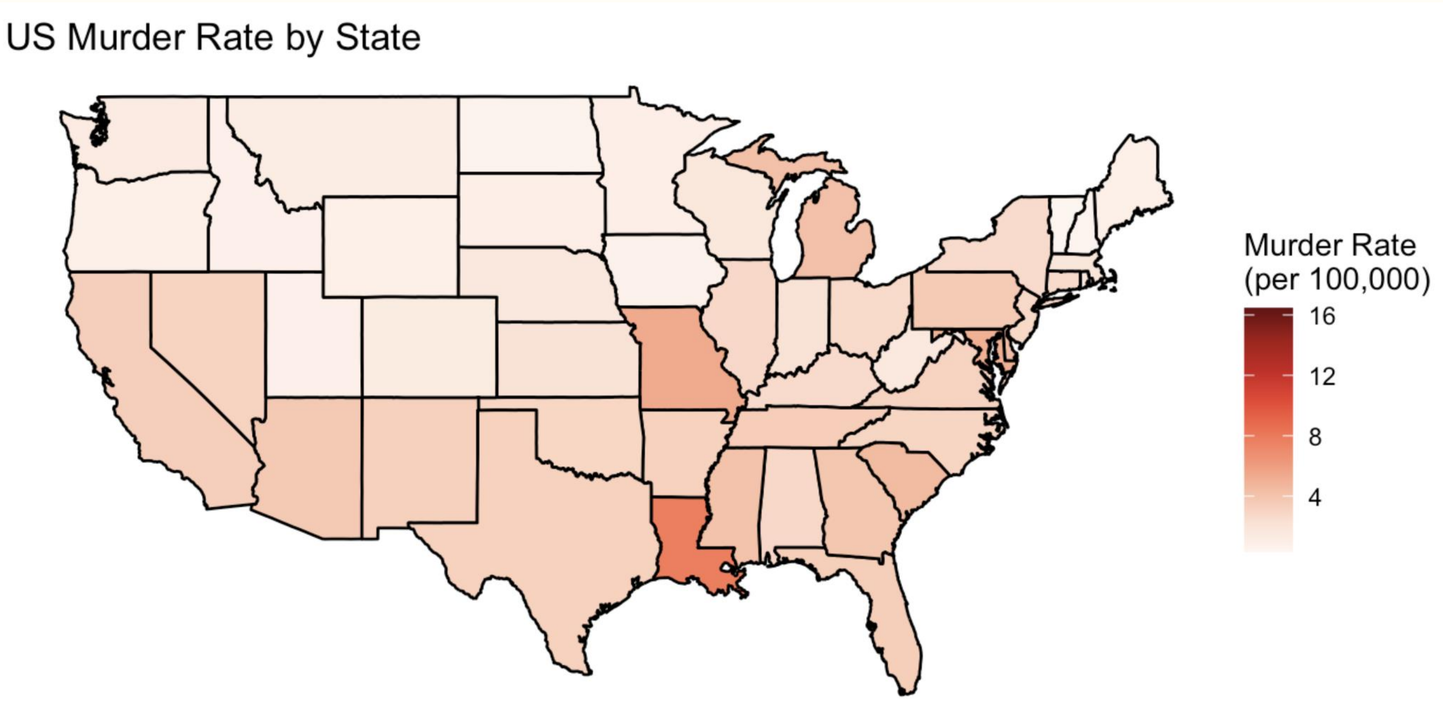
```
##        state abb     region population total     rate
## 1    Alabama  AL    alabama    4779736   135 2.824424
## 2     Alaska  AK     alaska     710231    19 2.675186
## 3    Arizona  AZ    arizona    6392017   232 3.629527
## 4   Arkansas  AR   arkansas    2915918    93 3.189390
## 5 California  CA california   37253956  1257 3.374138
## 6   Colorado  CO   colorado    5029196    65 1.292453
```

```
merged_data <- left_join(states_map, murders, by = "region")
head(merged_data)
```

```
##        long      lat group order  region subregion   state abb population total
## 1 -87.46201 30.38968     1     1 alabama      <NA> Alabama  AL    4779736   135
## 2 -87.48493 30.37249     1     2 alabama      <NA> Alabama  AL    4779736   135
## 3 -87.52503 30.37249     1     3 alabama      <NA> Alabama  AL    4779736   135
## 4 -87.53076 30.33239     1     4 alabama      <NA> Alabama  AL    4779736   135
## 5 -87.57087 30.32665     1     5 alabama      <NA> Alabama  AL    4779736   135
## 6 -87.58806 30.32665     1     6 alabama      <NA> Alabama  AL    4779736   135
##       rate
## 1 2.824424
## 2 2.824424
## 3 2.824424
## 4 2.824424
## 5 2.824424
## 6 2.824424
```

Plotting state murder rates

```
merged_data %>%
ggplot(aes(long, lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  scale_fill_gradientn(colors = brewer.pal(9, "Reds")) +
  theme_void() +
  labs(title = "US Murder Rate by State",
       fill = "Murder Rate \n(per 100,000)") +
  coord_fixed(1.3)
```



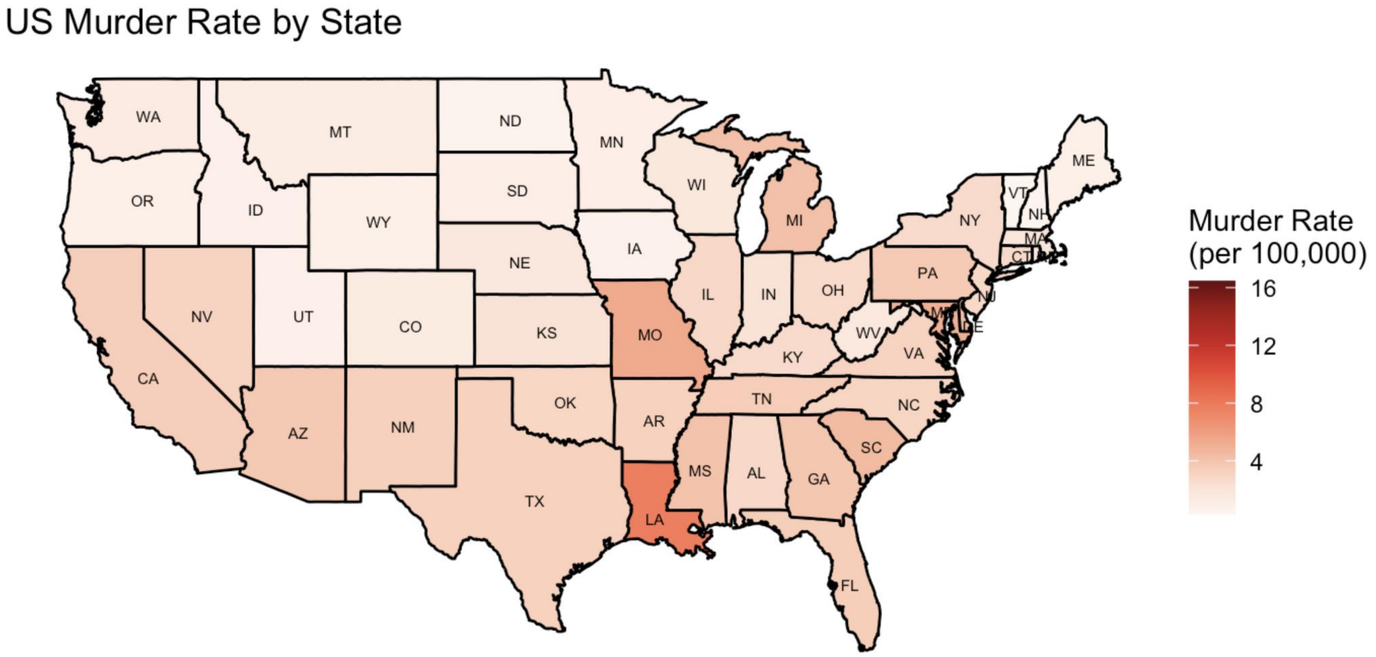
Adding state labels

```
library(ggrepel)
centroids <- data.frame(region = tolower(state.name),
                        long = state.center$x,
                        lat = state.center$y)

centroids$abb <- state.abb[match(centroids$region,
                                 tolower(state.name))]

centroids <- centroids %>%
  filter(!region %in% c("alaska", "hawaii"))

merged_data %>%
ggplot(aes(long, lat, group = group, fill = rate)) +
  geom_polygon(color = "black") +
  scale_fill_gradientn(colors = brewer.pal(9, "Reds")) +
  theme_void() +
  labs(title = "US Murder Rate by State",
       fill = "Murder Rate \n(per 100,000)") +
  coord_fixed(1.3) +
  with(centroids,
       annotate(geom = "text", x = long, y = lat, label = abb,
                size = 2, color = "black", family = "Arial"))
```



Summary

- `map_data("states")` can be used to create a map of the contiguous United States
- The `tolower` function makes all letters in a string lowercase
- `state.center$x` and `state.center$y` contain the centoid longitude and latitude points for each US state
- Centroid locations can be used to plot state labels